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Professor Stephen Scott
Head of the School of Mechanical Engineering
Faculty of Engineering and Physical Sciences
University of Leeds
Leeds LS2 9JT, United Kingdom

Dear Professor Scott and Members of the Selection Committee,

Application for Lecturer in Sustainable Engineering, Reference EPSME1208

I am writing to apply for the post of **Lecturer in Sustainable Engineering** in the School of Mechanical Engineering. My research field is the multi-physics design, thermal management and AI-assisted modelling of electrified powertrains, which sits directly on the post's focus on **sustainable engineering with regard to automotive engineering applications**. I am currently a Research Assistant at the Power Electronics and Machines Centre (PEMC), University of Nottingham, having completed my PhD at Southeast University (SEU), Nanjing, in September 2025 with the dissertation *Multi-physics Analysis and Design of Hairpin Winding PM Traction Machines in Electric Vehicles*. The PhD is in Electrical Engineering, a closely related discipline, and its substance is as much mechanical as electrical: lumped-parameter thermal networks and fluid-solid coupled oil-cooling models, mechanical stress and structural-modal analysis of high-speed rotors, and manufacturing-tolerance design, all validated on AVL dynamometers.

My research record includes **22 journal papers (17 in IEEE Transactions)**, ten as first author, 18 peer-reviewed conference papers and 9 patents, recognised by the **2025 IEEE TEC Electric Vehicle Design Competition (1st Prize)** and the **2025 IEEE Transactions on Transportation Electrification Prize Paper Award (2nd Prize)**. Through a multi-year collaboration with Inovance, a world-leading EV traction-machine supplier, I delivered two high-speed oil-cooled PM traction machines for A0-class and A-class vehicles (18,000 rpm, 113 kW; 20,000 rpm, 270 kW) through the full industrial cycle from multi-physics design and engineering-drawing release to dynamometer characterisation.

The post asks for a coherent and ambitious research vision in sustainable engineering. Mine is a programme on **AI-driven design and digital twins of electrified powertrains** for net-zero road transport, built from three connected strands: multi-physics modelling of automotive traction machines and their converters; **deep-learning surrogate models** that replace expensive finite-element evaluations, where my ICEM 2026 paper reaches $R^2 = 0.9996$ for torque prediction with more than 94% inference speed-up, built on an in-house Python finite-element code package; and experimental validation on dynamometer-tested industrial prototypes. The programme serves sustainability twice over, through vehicles that consume less energy in operation and through magnet-free and reduced-rare-earth machine topologies that lower material criticality, the focus of my current work on the multi-institution AIMD programme (Nottingham, Bristol, Newcastle). I would pursue an **EPSRC New Investigator Award** within the first eighteen months, Innovate UK and **Advanced Propulsion Centre** industrially co-funded projects building on my prototype delivery record, and participation in larger consortium bids as the group grows. Within the School I see natural links to the **Institute of Thermofluids** on thermal management of electrified powertrains and to the **Institute of Design, Robotics and Manufacturing** on design optimisation and electric drives as actuation.

On teaching, I served as teaching assistant on the SEU postgraduate course *Integrated Practice of Electrical Machine Systems*, covering tutorial delivery, daily consultation hours, structured written feedback on student project proposals and mark recommendations. I have research-derived materials ready to develop into **research-led teaching** that maps onto the School's programmes, including the **Automotive Engineering MEng/BEng** with its coverage of electric and hybrid vehicle technology and modern drivetrain technology: dynamometer data from two industrial automotive prototypes as worked examples, the Python finite-element package for simulation-based design coursework and a digital-twin exercise, and embedded motor-drive laboratory practicals on TI C2000 and STM32 platforms supporting mechatronics, dynamics and control. I would be glad to support Leeds Gryphon Racing on the electric powertrain side, would act as a personal tutor, and would work toward **Fellowship of the Higher Education Academy** as an early priority.

Having completed my BEng, MSc and PhD at Southeast University, I know the Chinese university system from the inside and am a native Mandarin speaker, so I am well placed to support the School's **transnational education** activities, including overseas teaching of up to six weeks per academic year. I have studied and

worked across distinctly different institutional cultures in China and the United Kingdom, and I am committed to inclusive teaching, supervision and mentoring practice and to the wider academic citizenship of the School.

My academic CV, including a full publication list, is attached, and my responses to the selection criteria are provided in the supporting statement section of the online application. I would be glad to discuss my application further at interview.

Yours sincerely,

Wentao Zhang